

Safety Features — Original File Protection

Overview

QB-TimeWarp implements **multiple layers of protection** to ensure that original QuickBooks company files on the Desktop are **NEVER modified**. All migration operations work exclusively with copies in dedicated Working directories.

Folder Structure

```
Desktop (READ-ONLY originals – NEVER touched):
├── Joshua's Gold Coast\      ← QB 2023 source (30MB, testing file)
│   └── *.qbw
├── Blank Template\          ← QB 2021 target
│   └── *.qbw
└── Air Masters\             ← QB 2023 production (360MB – DO NOT USE)
    └── Air-Masters-QB-2023.qbw

C:\QB-TimeWarp\ (Application workspace):
├── Working\
│   ├── Source\              ← COPY of Joshua's Gold Coast .qbw
│   └── Target\               ← COPY of Blank Template .qbw
├── ExportedData\
├── Schemas\
├── Logs\
└── Validation\
```

Safety Layers

Layer 1: WorkingDirectoryManager (Pre-Operation)

- **When:** Before ANY QuickBooks operations begin
- **What:** Copies original files from Desktop to `Working\Source\` and `Working\Target\`
- **Verification:** Compares file sizes after copy to ensure integrity
- **Skip logic:** Won't overwrite existing working copies unless `--refresh` is used
- **Service:** `Services/WorkingDirectoryManager.cs`

Layer 2: Configuration Path Override (Program.cs)

- **When:** After working copies are created, before any service initialization
- **What:** Overwrites `QuickBooks.QB2023.CompanyFilePath` and `QuickBooks.QB2021.CompanyFilePath` to point to Working directory copies
- **Result:** All downstream services automatically receive working copy paths

Layer 3: QBConnectionManager Validation (Every Connection)

- **When:** Every time a QB connection is created AND every time `.Connect()` is called
- **What:** Checks the company file path against a list of protected Desktop folder patterns
- **Action:** Throws `OriginalFileProtectionException` if a protected path is detected
- **Protected patterns:**

- `\Desktop\Joshua's Gold Coast`
- `\Desktop\Blank Template`
- `\Desktop\Air Masters`
- `\Desktop\AirMasters`

Layer 4: Desktop Path Rejection (Program.cs Fallback)

- **When:** If `AutoCreateWorkingCopies` is disabled
- **What:** Validates that configured `CompanyFilePath` values don't point to Desktop folders
- **Action:** Throws `OriginalFileProtectionException` with clear instructions

Layer 5: Working Copy Validation Helper

- **What:** `WorkingDirectoryManager.ValidateNotOriginalPath()` method
- **Usage:** Can be called before any write/modify operation to verify the target path
- **Checks:** Against configured Desktop source/target folders AND general Desktop path patterns

Configuration

In `appsettings.json` :

```
{
  "SourceFiles": {
    "DesktopFolder": "C:\\Users\\AIAgent\\Desktop\\Joshua's Gold Coast",
    "CompanyFileName": "*.qbw"
  },
  "TargetFiles": {
    "DesktopFolder": "C:\\Users\\AIAgent\\Desktop\\Blank Template",
    "CompanyFileName": "*.qbw"
  },
  "WorkingDirectories": {
    "SourcePath": "C:\\QB-TimeWarp\\Working\\Source",
    "TargetPath": "C:\\QB-TimeWarp\\Working\\Target",
    "AutoCreateWorkingCopies": true,
    "PreserveOriginals": true
  }
}
```

Command-Line Options

Option	Description
<code>--refresh</code>	Force re-copy originals to Working directories (deletes existing copies first)
<code>--cleanup</code>	Remove all Working directories and exit (originals remain untouched)

Exit Codes

Code	Meaning
0	Success
1	General error
2	Working copy creation/validation failure
99	SAFETY VIOLATION — attempted operation on original file

What Happens If...

...working copies already exist?

- They are reused without re-copying (faster startup)
- Use `--refresh` to force a fresh copy from originals

...an original file path is accidentally configured?

- `QBConnectionManager` blocks the connection attempt
- `OriginalFileProtectionException` is thrown
- Application exits with code 99
- Error message clearly identifies which file and why

...the Desktop folders don't exist?

- `WorkingDirectoryManager` throws `WorkingCopyException`
- Application exits with code 2
- Error message identifies which folder is missing

...the .qbw file can't be found?

- Glob pattern `*.qbw` is used to find the company file
- If no match, a clear error identifies the folder and pattern
- If multiple matches, the first is used with a warning

Logging

At startup, you'll see prominent log messages:

```
WORKING WITH COPIES – ORIGINALS PRESERVED
```

```
Source (QB 2023): C:\QB-TimeWarp\Working\Source\*.qbw
Target (QB 2021): C:\QB-TimeWarp\Working\Target\*.qbw
```

And for each `QBConnectionManager`:

[QB2023-Export] ✓ Using working copy (originals protected): C:\QB-Time-Warp\Working\Source*.qbw